

Auteur: Mark Gijsbrechts
Studierichting: Informatica
Oefeningenreeks 5K nr. 1.4

$$r = \sin^2 \frac{\theta}{2} \text{ op het interval } [0, 2\pi]$$

$$r^2 = \sin^4 \frac{\theta}{2}$$

$$r' = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \cdot \frac{1}{2} = \sin \frac{\theta}{2} \cos \frac{\theta}{2}$$

$$(r')^2 = \sin^2 \frac{\theta}{2} \cos^2 \frac{\theta}{2}$$

$$L = \int_0^{2\pi} \sqrt{\sin^4 \frac{\theta}{2} + \sin^2 \frac{\theta}{2} \cos^2 \frac{\theta}{2}} d\theta$$

$$= \int_0^{2\pi} \sqrt{\sin^2 \frac{\theta}{2} \left(\sin^2 \frac{\theta}{2} + \cos^2 \frac{\theta}{2} \right)} d\theta$$

$$= \int_0^{2\pi} \sqrt{\sin^2 \frac{\theta}{2}} d\theta$$

$$= \int_0^{2\pi} \sin \frac{\theta}{2} d\theta$$

$$u = \frac{\theta}{2} \Rightarrow du = \frac{1}{2} d\theta \Rightarrow d\theta = 2du$$

$$= 2 \int_0^{\pi} \sin u du$$

$$= -2 [\cos u]_0^{\pi}$$

$$= -2 [-1 - 1]$$

$$= 4$$

References



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